How-To Library

Informatica

Importing and Exporting Data Assets Between Oracle and Enterprise Data Preparation

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Abstract

You can import data assets from an Oracle database to your data lake, and can export data assets from your data lake to an Oracle database. Review this article to prepare Enterprise Data Preparation to import and export data assets.

Supported Versions

- Enterprise Data Lake 10.2 x
- Enterprise Data Lake 10.4.0

Table of Contents

Overview. 2
Download the JDBC Driver. 2
Create JDBC Connections to Access an Oracle Database. 3
Configure the sqoopSchemaUpperCase Custom Option. 4
Limitations for Oracle Import and Export Operations. 5

Overview

You can import data from Oracle database tables into your data lake. You can also export data from your data lake to an Oracle database.

After you import a data asset from an Oracle database, you can add the imported data as a worksheet to a project in Enterprise Data Preparation. After you prepare an asset and export the data to an Oracle database, you can perform further operations on the data in an Oracle database.

To prepare to import and export data assets between an Oracle database and Enterprise Data Preparation, complete the following steps:

1. Download the Oracle JDBC driver that corresponds to your Oracle database release, and then copy the driver to the Data Integration Service machine associated with Enterprise Data Preparation.
2. Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.
3. If you export data from a data lake on a Cloudera CDH cluster, add the sqoopSchemaUpperCase custom property to the Enterprise Data Preparation Service.

Download the JDBC Driver

Download the Oracle JDBC driver that corresponds to your Oracle database release, and then copy the driver to the Data Integration Service machine associated with Enterprise Data Preparation.

When you import or export data, the Data Integration Service uses Sqoop to connect to the cluster, and then executes a Sqoop job on the cluster to import or export the data. Sqoop uses the driver to connect to the Oracle database target when the import or export job runs on the cluster.

You can download the Oracle JDBC driver .jar file from the following link:

http://www.oracle.com/technetwork/database/application-development/jdbc/downloads/index.html

Copy the .jar file to the following directory on the machine where the Data Integration Service runs:

```
<Informatica installation directory>/externaljdbcjars
```

Create JDBC Connections to Access an Oracle Database

Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.

You include the arguments Sqoop uses to connect to the Oracle database in each JDBC connection.

1. In the Administrator tool, click the **Manage** tab > **Connections** view.
2. In the Domain Navigator, click **Actions > New > Connection**.

   The New Connection dialog box appears.
3. In the New Connection dialog box, select **Database** > **JDBC**.
4. Click **OK**.

   The **New Connection** wizard appears.
5. Enter the JDBC connection properties required to connect to the Oracle database.

   The following table describes the connection properties you enter:

Property	Description
Name	Name of the connection. The name is not case sensitive, and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Informatica services use to identify the connection. The string is not case sensitive. The string must be 255 characters or less, and must be unique within the domain. Default value is the connection name.
Description	Description of the connection. The description cannot exceed 765 characters.
User Name	Database user name.
Password	Password for the database user name.
JDBC Driver Class Name	Name of the JDBC driver class packaged with Informatica. Enterprise Data Preparation uses the driver to connect to the Oracle database. Enter the following value: `com.informatica.jdbc.oracle.OracleDriver`

Connection String	Connection string to connect to the database. Use the following format for the connection string:
	`jdbc:informatica:oracle://<database host name>:<port>;SID=<database name>`
	The following example shows a sample connection string:
	`jdbc:informatica:oracle://db_server_01:1521;SID=mydatabase`

Environment SQL	Optional. SQL commands to set the database environment when you connect to the target. The Data Integration Service executes the SQL each time it connects to the database.
Transaction SQL	Optional. SQL commands to set the database environment when you connect to the database. The Data Integration Service executes the SQL at the beginning of each transaction.

Property	Description
SQL Identifier Character	Select " " from the drop down list to connect to the database from a Hortonworks HDP cluster or from a Cloudera CDH cluster.
Support Mixed-case Identifiers	Select **Yes** from the drop down list.
Use Sqoop Connector	Select **Sqoop v1.x** to enable Sqoop connectivity.

Sqoop Arguments	Sqoop arguments required to connect to the database.
	Specify the connection string Sqoop uses to connect to the database in the `--connect` argument. Specify the class name of the Oracle JDBC driver Sqoop uses in the `--driver` argument.
	Use the following format for the argument string:
	`--connect jdbc:oracle:thin:@<database host name>:<port>:<database name> --driver <JDBC driver name>`
	The following example shows a sample argument string:
	`--connect jdbc:oracle:thin:@db_server_01:1521:mydatabase --driver oracle.jdbc.driver.OracleDriver`

6. When you finish entering the connection properties, click **Test Connection** to test the connection.
7. Click **Finish**.

Configure the sqoopSchemaUpperCase Custom Option

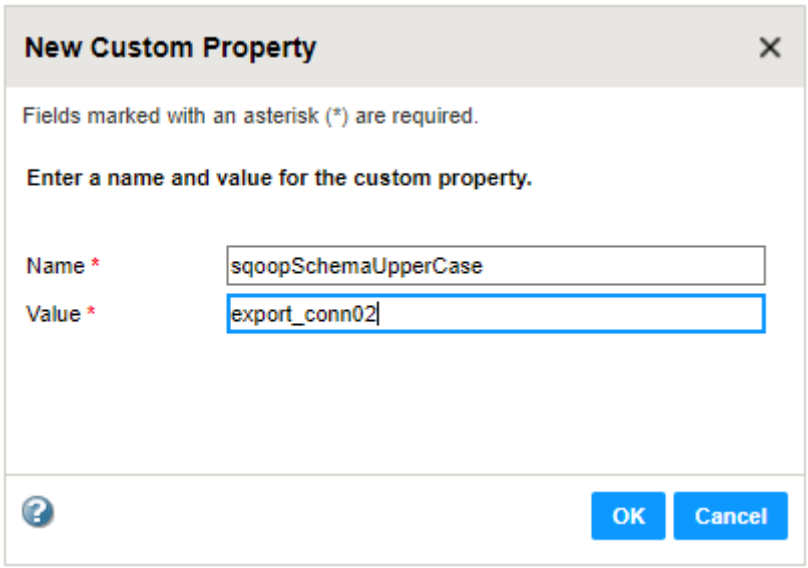
If you export data to an Oracle database from a data lake on a Cloudera CDH cluster, use the Administrator tool to add the sqoopSchemaUpperCase custom property to the Enterprise Data Preparation Service. Enter the name of each JDBC connection that Enterprise Data Preparation uses to export data as the custom property value.

Cloudera CDH requires the use of uppercase table names and column names. By default, Enterprise Data Preparation uses lowercase names for tables and columns. The sqoopSchemaUpperCase custom property forces Enterprise Data Preparation to convert all table names and column names to uppercase.

Note: You do not need to add the custom property to connect to an Oracle database from any other distribution.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. Click the **Properties** view.
4. Click the pencil icon in the Custom Options section.
5. Click **New** in the Edit Custom Options dialog box.
6. Enter sqoopSchemaUpperCase as the custom property name.
7. Enter the name of each JDBC connection Enterprise Data Preparation uses to import and export data as the custom property value. If you enter multiple connection names, separate each connection name with a comma.

The following image shows the **New Custom Property** dialog box:



8. Click **OK**.

You can now import and export data assets between Enterprise Data Preparation and an Oracle database.

For more information about importing and exporting data assets, see the "Import, Export, and Upload Data" chapter in the Enterprise Data Preparation User Guide.

Limitations for Oracle Import and Export Operations

Import and export operations between Enterprise Data Preparation and an Oracle database might fail because of third-party limitations.

Import operation limitations

Import operations from Oracle might fail because of the following third-party limitations:

- The table contains a column of timestamp data type with time zone.
- The table contains a column with a negative scale.

Export operation limitations

Export operations to Oracle might fail because of the following third-party limitations:

- The data asset contains a column of float data type.
- The data asset contains a column of varchar data type.
- The data asset contains a column of double data type.

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